C4 Solutions: trigonometric functions

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All exact values and equation solutions were checked numerically with Python; identities were checked with SymPy.

Practice

1a. $135° = \frac{135}{180}\pi = \frac{3\pi}{4}$; $210° = \frac{7\pi}{6}$; $330° = \frac{11\pi}{6}$. **1b.** $\frac{5\pi}{3} \cdot \frac{180°}{\pi} = 300°$; $\frac{7\pi}{12} \cdot \frac{180°}{\pi} = 105°$. **1c.** $\frac{180°}{\pi} \approx 57.3°$.

2.

a. $\frac{5\pi}{6} = \pi - \frac{\pi}{6}$, quadrant II, sin > 0: $\frac{1}{2}$.
b. $\frac{4\pi}{3} = \pi + \frac{\pi}{3}$, quadrant III, cos < 0: $-\frac{1}{2}$.
c. tan is odd: $-\tan\frac{\pi}{4} = -1$.
d. $\frac{7\pi}{4} = 2\pi - \frac{\pi}{4}$, quadrant IV, sin < 0: $-\frac{1}{2}\sqrt{2}$.
e. $\frac{11\pi}{6} = 2\pi - \frac{\pi}{6}$, quadrant IV, cos > 0: $\frac{1}{2}\sqrt{3}$.
f. $\frac{2\pi}{3} = \pi - \frac{\pi}{3}$, quadrant II, tan < 0: $-\sqrt{3}$.

3a. Equilibrium $y = 3$, amplitude 2, period $\frac{2\pi}{4} = \frac{\pi}{2}$, starting point $\left(\frac{\pi}{8}, 3\right)$.

3b. Maximum $3 + 2 = 5$, a quarter period after the start: $x = \frac{\pi}{8} + \frac{\pi}{8} = \frac{\pi}{4}$. Minimum $3 - 2 = 1$, three quarters of a period after the start: $x = \frac{\pi}{8} + \frac{3\pi}{8} = \frac{\pi}{2}$. (Also at $x = \frac{\pi}{2} - \frac{\pi}{2} = 0$, but the question asks for $x > 0$.)

3c. On $[0, \pi]$ (two full periods): minimum at 0, middle going up at $\frac{\pi}{8}$, maximum at $\frac{\pi}{4}$, middle going down at $\frac{3\pi}{8}$, minimum at $\frac{\pi}{2}$, and the same pattern again up to a minimum at π. Check with GeoGebra.

4. $\sin x = -\frac{1}{2} = \sin\left(-\frac{\pi}{6}\right)$. $x = -\frac{\pi}{6} + k2\pi$ or $x = \pi - \left(-\frac{\pi}{6}\right) + k2\pi = \frac{7\pi}{6} + k2\pi$. In $[0, 2\pi]$: $x = \frac{7\pi}{6}$ and $x = \frac{11\pi}{6}$.

5. $\cos 2x = \cos\frac{5\pi}{6}$. So $2x = \frac{5\pi}{6} + k2\pi$ or $2x = -\frac{5\pi}{6} + k2\pi$, i.e. $x = \frac{5\pi}{12} + k\pi$ or $x = -\frac{5\pi}{12} + k\pi$. In $[0, 2\pi]$: $\frac{5\pi}{12}$, $\frac{17\pi}{12}$ and $\frac{7\pi}{12}$, $\frac{19\pi}{12}$.

6a. $\tan x = \tan\frac{\pi}{3}$, so $x = \frac{\pi}{3} + k\pi$. **6b.** $\tan 2x = \tan\left(-\frac{\pi}{4}\right)$, so $2x = -\frac{\pi}{4} + k\pi$, $x = -\frac{\pi}{8} + k \cdot \frac{\pi}{2}$.

7. Arc: $r\theta = 6 \cdot \frac{2\pi}{3} = 4\pi$ cm. Sector: $\frac{1}{2}r^2\theta = \frac{1}{2} \cdot 36 \cdot \frac{2\pi}{3} = 12\pi$ cm².

8a. $(\sin x + \cos x)^2 = \sin^2 x + 2\sin x\cos x + \cos^2 x = 1 + \sin 2x$.

8b. $1 - \cos 2x = 1 - (1 - 2\sin^2 x) = 2\sin^2 x$ and $\sin 2x = 2\sin x\cos x$. So the quotient is $\dfrac{2\sin^2 x}{2\sin x\cos x} = \dfrac{\sin x}{\cos x} = \tan x$ (we may cancel $\sin x$, since $\sin 2x \neq 0$ implies $\sin x \neq 0$ and $\cos x \neq 0$).

9. $\gamma = 180° - 40° - 65° = 75°$. Sine rule: $b = \dfrac{a\sin\beta}{\sin\alpha} = \dfrac{10\sin 65°}{\sin 40°} \approx 14.10$ and $c = \dfrac{10\sin 75°}{\sin 40°} \approx 15.03$.

10. The largest angle is opposite the longest side, 13. Cosine rule: $\cos\gamma = \dfrac{7^2+8^2-13^2}{2\cdot 7\cdot 8} = \dfrac{49+64-169}{112} = \dfrac{-56}{112} = -\dfrac{1}{2}$. So $\gamma = \frac{2\pi}{3}$ (120°).

11. Equilibrium $a = \frac{7+1}{2} = 4$, amplitude $b = \frac{7-1}{2} = 3$. Half a period is $4-1=3$, so period 6 and $c = \frac{2\pi}{6} = \frac{\pi}{3}$. Cosine form (maximum at $x=1$): $y = 4+3\cos\left(\frac{\pi}{3}(x-1)\right)$. Sine form: the starting point is a quarter period (1.5) before the maximum, at $x = -\frac{1}{2}$: $y = 4+3\sin\left(\frac{\pi}{3}\left(x+\frac{1}{2}\right)\right)$.

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12. $\cos 2x = 1-2\sin^2 x$, so $\sin x = 1-2\sin^2 x$, i.e. $2\sin^2 x+\sin x-1=0$, $(2\sin x-1)(\sin x+1)=0$. $\sin x = \frac{1}{2}$: $x = \frac{\pi}{6}$ or $x = \frac{5\pi}{6}$. $\sin x = -1$: $x = \frac{3\pi}{2}$. **Answer:** $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$.

13. $\sin A = \sin B$ with $A = 2x$, $B = x+\frac{\pi}{3}$:

- $2x = x+\frac{\pi}{3}+k2\pi \Rightarrow x = \frac{\pi}{3}+k2\pi$;
- $2x = \pi - x - \frac{\pi}{3}+k2\pi \Rightarrow 3x = \frac{2\pi}{3}+k2\pi \Rightarrow x = \frac{2\pi}{9}+k\frac{2\pi}{3}$.

In $[0,2\pi]$: $x = \frac{\pi}{3}$, and $x = \frac{2\pi}{9}, \frac{8\pi}{9}, \frac{14\pi}{9}$.

14. $\cos\frac{\pi}{12} = \cos\left(\frac{\pi}{3}-\frac{\pi}{4}\right) = \cos\frac{\pi}{3}\cos\frac{\pi}{4}+\sin\frac{\pi}{3}\sin\frac{\pi}{4} = \frac{1}{2}\cdot\frac{1}{2}\sqrt{2}+\frac{1}{2}\sqrt{3}\cdot\frac{1}{2}\sqrt{2} = \dfrac{\sqrt{2}+\sqrt{6}}{4}$.

15a. $2\sin\left(x+\frac{\pi}{3}\right) = 2\left(\sin x\cos\frac{\pi}{3}+\cos x\sin\frac{\pi}{3}\right) = 2\left(\frac{1}{2}\sin x+\frac{1}{2}\sqrt{3}\cos x\right) = \sin x+\sqrt{3}\cos x$.

15b. The maximum of $2\sin(\dots)$ is 2.

15c. $\sin\left(x+\frac{\pi}{3}\right) = \frac{1}{2} = \sin\frac{\pi}{6}$: $x+\frac{\pi}{3} = \frac{\pi}{6}+k2\pi \Rightarrow x = -\frac{\pi}{6}+k2\pi$; or $x+\frac{\pi}{3} = \frac{5\pi}{6}+k2\pi \Rightarrow x = \frac{\pi}{2}+k2\pi$. In $[0,2\pi]$: $x = \frac{\pi}{2}$ and $x = \frac{11\pi}{6}$.

16a. Period 12.4 hours. The starting point is at $t=2$. High tide a quarter period later: $t = 2+3.1 = 5.1$, i.e. 05:06. Low tide three quarters of a period after the start: $t = 2+9.3 = 11.3$, i.e. 11:18. (Before $t=5.1$ there is no low tide after midnight: the previous one was at $11.3-12.4<0$.)

16b. $h(t) = 165 \iff \sin\left(\frac{2\pi}{12.4}(t-2)\right) = \frac{1}{2}$. Write $u = \frac{2\pi}{12.4}(t-2)$. In one period starting at $u=0$, $\sin u > \frac{1}{2}$ for $\frac{\pi}{6} < u < \frac{5\pi}{6}$, an interval of length $\frac{2\pi}{3}$, which is one third of the period 2π. So per tidal period the water is above 165 cm for $\frac{1}{3}\cdot 12.4 = \frac{62}{15}$ hours $= 4\frac{2}{15}$ hours = **4 hours 8 minutes**. (In terms of t: from $t = 2+\frac{12.4}{12} \approx 3.03$ to $t = 2+\frac{5\cdot 12.4}{12} \approx 7.17$.)

17a. $\cos 3x = \cos(2x+x) = \cos 2x\cos x - \sin 2x\sin x = (2\cos^2 x-1)\cos x - 2\sin^2 x\cos x$. Use $\sin^2 x = 1-\cos^2 x$: $= 2\cos^3 x - \cos x - 2\cos x + 2\cos^3 x = 4\cos^3 x - 3\cos x$. □

17b. With $x = \frac{\pi}{9}$: $\cos\frac{\pi}{3} = \frac{1}{2} = 4c^3-3c$. Multiply by 2: $8c^3-6c-1=0$. □

18. $\angle ACB = 180°-52°-71° = 57°$. Sine rule in triangle ABC: $AC = \dfrac{500\sin 71°}{\sin 57°} \approx 563.7$ m. The distance to the coast is the altitude from C to line AB: $AC\sin 52° \approx 444.2$ m. **About 444 m.**

19a. Area $= \frac{1}{2}bc\sin\alpha = \frac{1}{2}\cdot 6\cdot 10\cdot\sin 150° = 30\cdot\frac{1}{2} = 15$.

19b. $a^2 = b^2+c^2-2bc\cos\alpha = 36+100-120\cdot\left(-\frac{1}{2}\sqrt{3}\right) = 136+60\sqrt{3}$. So $a = \sqrt{136+60\sqrt{3}} \approx 15.49$.

20a. From $\cos 2x = 1 - 2\sin^2 x$: $\sin^2 x = \dfrac{1-\cos 2x}{2} = \frac{1}{2} - \frac{1}{2}\cos 2x$.

20b. Equilibrium $\frac{1}{2}$, amplitude $\frac{1}{2}$, period $\frac{2\pi}{2} = \pi$.

21. Put $u = \cos x$: $2u^2 - 3u + 1 = 0$, $(2u-1)(u-1) = 0$, $u = \frac{1}{2}$ or $u = 1$. $\cos x = \frac{1}{2}$: $x = \frac{\pi}{3}, \frac{5\pi}{3}$. $\cos x = 1$: $x = 0, 2\pi$. **Answer:** $x = 0, \frac{\pi}{3}, \frac{5\pi}{3}, 2\pi$.

Challenge

22a. $OA = OB = 1$ and $\angle AOB = \alpha - \beta$ (which lies in $(0, \pi)$, so OAB is a real triangle). Cosine rule: $AB^2 = 1 + 1 - 2\cos(\alpha - \beta) = 2 - 2\cos(\alpha - \beta)$.

22b. $AB^2 = (\cos\alpha - \cos\beta)^2 + (\sin\alpha - \sin\beta)^2 = (\cos^2\alpha + \sin^2\alpha) + (\cos^2\beta + \sin^2\beta) - 2(\cos\alpha\cos\beta + \sin\alpha\sin\beta) = 2 - 2(\cos\alpha\cos\beta + \sin\alpha\sin\beta)$. Compare with (a): $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$. □

22c. The cosine rule is about angles of a triangle, which lie strictly between 0 and π. So this proof only covers $0 < \alpha - \beta < \pi$ (and, by symmetry, $-\pi < \alpha - \beta < 0$). Values like $\alpha - \beta = 0$, π or $\frac{5\pi}{2}$ need a separate argument (periodicity, or degenerate cases). The rotation proof in the notes works for all real α, β at once.

23. *Case α acute.* Let BD be the diameter through B, so $BD = 2R$. If $D = C$, then BC is a diameter, $\alpha = 90°$, which is not acute; so $D \neq C$. Since α is acute, A and D lie on the same arc cut off by chord BC, so by the inscribed angle theorem $\angle BDC = \angle BAC = \alpha$. Angle $\angle BCD = 90°$ (Thales: angle in a semicircle). In the right triangle BCD: $\sin\angle BDC = \dfrac{BC}{BD}$, i.e. $\sin\alpha = \dfrac{a}{2R}$. So $\dfrac{a}{\sin\alpha} = 2R$. *Case $\alpha = 90°$:* BC is a diameter, $a = 2R$ and $\sin\alpha = 1$. True. *Case α obtuse:* A and D lie on opposite arcs, so $ABDC$ is a cyclic quadrilateral and $\angle BDC = 180° - \alpha$. Since $\sin(180° - \alpha) = \sin\alpha$, the same computation works. □

24a. $5\theta = \frac{\pi}{2}$, so $2\theta = \frac{\pi}{2} - 3\theta$ and $\sin 2\theta = \sin\left(\frac{\pi}{2} - 3\theta\right) = \cos 3\theta$.

24b. $2\sin\theta\cos\theta = 4\cos^3\theta - 3\cos\theta$. Divide by $\cos\theta \neq 0$: $2\sin\theta = 4\cos^2\theta - 3 = 4(1 - \sin^2\theta) - 3 = 1 - 4\sin^2\theta$. So $4\sin^2\theta + 2\sin\theta - 1 = 0$.

24c. Quadratic in $s = \sin\theta$: $s = \dfrac{-2 \pm \sqrt{4+16}}{8} = \dfrac{-1 \pm \sqrt{5}}{4}$. Since $0 < \theta < \frac{\pi}{2}$, $s > 0$, so $\sin\frac{\pi}{10} = \dfrac{\sqrt{5}-1}{4}$. □ (This number is $\frac{1}{2\varphi}$, with φ the golden ratio; it is why the regular pentagon is full of golden ratios.)

25a. $\tan(\alpha + \beta) = \dfrac{\sin\alpha\cos\beta + \cos\alpha\sin\beta}{\cos\alpha\cos\beta - \sin\alpha\sin\beta}$. Divide numerator and denominator by $\cos\alpha\cos\beta$ (nonzero, since the tangents are defined): $= \dfrac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$. □

25b. Let $\alpha = \arctan\frac{1}{2}$ and $\beta = \arctan\frac{1}{3}$. Then $\tan(\alpha+\beta) = \dfrac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{6}} = \dfrac{5/6}{5/6} = 1$. Both α and β lie in $\left(0, \frac{\pi}{4}\right)$ because $0 < \frac{1}{3} < \frac{1}{2} < 1 = \tan\frac{\pi}{4}$ and tan is increasing there. So $0 < \alpha + \beta < \frac{\pi}{2}$, and the only angle in that interval with tangent 1 is $\frac{\pi}{4}$. □

26. Area $F = \frac{1}{2}ab\sin\gamma$, so $16F^2 = 4a^2b^2\sin^2\gamma = 4a^2b^2(1-\cos\gamma)(1+\cos\gamma) = (2ab - 2ab\cos\gamma)(2ab + 2ab\cos\gamma)$. Cosine rule: $2ab\cos\gamma = a^2 + b^2 - c^2$. So

$$\begin{aligned} 16F^2 &= (2ab - a^2 - b^2 + c^2)(2ab + a^2 + b^2 - c^2) = (c^2 - (a-b)^2)((a+b)^2 - c^2) \\ &= (c - a + b)(c + a - b)(a + b - c)(a + b + c). \end{aligned}$$

With $a + b + c = 2s$: $c - a + b = 2(s - a)$, $c + a - b = 2(s - b)$, $a + b - c = 2(s - c)$. So $16F^2 = 16\,s(s-a)(s-b)(s-c)$ and $F = \sqrt{s(s-a)(s-b)(s-c)}$. □